Math 3
10-? Solve For Y

Name:_____
Date:_____ Per: ___

Notes:

1) $3x + y = 12$

2) $36 = 12x - 9y$

3) $4x + \frac{1}{3}y = 17$

4) $13x - \frac{1}{4}y = 3(x + 7)$

5) $12x + 10y + 13 = 6 + 7x + 4y$

Practice/You Try:

1) $2x + y = 7$

$-2x \quad -2x$

$y = -2x + 7$

2) $5x + 4y = 20$

$-5x \quad -5x$

$\frac{4y}{4} = \frac{-5x + 20}{4}$

$y = \frac{-5}{4}x + 5$

3) $12 = 9x + 3y$

$-9x \quad -9x$

$\frac{-9x + 12}{3} = \frac{3y}{3}$

$-3x + 4 = y$

$y = -3x + 4$

4) $18x - 2y = 26$

$-18x \quad -18x$

$\frac{-2y}{-2} = \frac{-18x + 26}{-2}$

$y = 9x - 13$

5) $14 = 7y - 6x$

$+6x \quad +6x$

$\frac{6x + 14}{7} = \frac{7y}{7}$

$\frac{6}{7}x + 2 = y$

$y = \frac{6}{7}x + 2$

6) $8x - 8y = 16$

$-8x \quad -8x$

$\frac{-8y}{-8} = \frac{-8x+16}{-8}$

$y = x - 2$

7) $124 = 4x - 10y$

$-4x \quad -4x$

$\frac{-4x+124}{-10} = \frac{-10y}{-10}$

$\frac{2}{5}x - \frac{62}{5} = y$

$y = \frac{2}{5}x - \frac{62}{5}$

8) $12 + \frac{1}{6}y = 13x$

$-12 \quad -12$

$6 \cdot \left(\frac{1}{6}y\right) = (13x - 12)6$

$y = 78x - 72$

9) $4x + 2y = 8(x + 10) - 2x$

$4x + 2y = 8x + 80 - 2x$

$4x + 2y = 6x + 80$

$-4x \quad -4x$

$\frac{2y}{2} = \frac{2x + 80}{2}$

$y = x + 40$

10) $9x + 10(y + 1) - 18 = 4 - 1(x + 3)$

$9x + 10y + 10 - 18 = 4 - x - 3$

$9x + 10y - 8 = -x + 1$

$-9x \quad -9x$

$10y - 8 = -10x + 1$

$+8 \quad +8$

$\frac{10y}{10} = \frac{-10x + 9}{10}$ → $y = -x + \frac{9}{10}$

11) $14x + 9y = 2y + 7(x - 11)$

$14x + 9y = 2y + 7x - 77$

$-2y \quad -2y$

$14x + 7y = 7x - 77$

$-14x \quad -14x$

$\frac{7y}{7} = \frac{-7x - 77}{7}$

$y = -x - 11$

12) $10(-6.5x + 28) = (2 - 1.3y)10$

$-65x + 280 = 20 - 13y$

$-20 \quad -20$

$\frac{-65x + 260}{-13} = \frac{-13y}{-13}$

$5x - 20 = y$

$y = 5x - 20$